VALENCY AND FORMULA 16

Atoms in chemical compounds have a characteristic combining power or **valency**. This involves the number of electrons in an incomplete outer shell and can be defined in several ways:

- It represents the charge on an ion
- It is equal to the number of electron pairs shared in a molecule
- It is the number of electrons transferred when forming an ion.

1 Use your own textbook to find an alternative definition for *valency*.

For elements in Groups I II, III, IV and 0, the group valency is the same as the group number (e.g. Group I elements have a valency of 1). For elements in Groups V, VI & VII there are two group valencies. One is the same as the group number and the other more likely valency is 8 minus the group number.

2 The Periodic Table involves groups and periods. Use examples of elements to explain the difference between groups and periods.

3 What is the most likely valency of a Group VI element? Explain your answer.

4 Learn the name, formula and charge on each ion shown in table 1.

+1 valency	+2 valency	+3 valency	+4 valency
lithium, Li^+	magnesium, Mg^{2+}	aluminium, Al^{3+}	tin (IV), Sn^{4+}
sodium, Na^+	calcium, Ca^{2+}	iron (III), Fe^{3+}	lead (IV), Pb^{4+}
potassium, K^+	barium, Ba^{2+}	chromium (III), Cr^{3+}	
copper (I), Cu^+	zinc, Zn^{2+}		
silver, Ag^+	copper (II), Cu^{2+}		
ammonium, NH_4^+*	mercury, Hg^{2+}		
	iron (II), Fe^{2+}		
	nickel (II), Ni^{2+}		
	tin (II), Sn^{2+}		
	lead (II), Pb^{2+}		

*The ammonium ion is not a metal ion, but has certain chemical similarities with metal ions.

Table 1

5 Name two things that the ions in table 1 have in common.

6 Give the formula for the ammonium ion.

7 What is the valency of the ammonium ion?

8 Give the two possible formulae for a copper ion.

The **transition metals** (also called the d-block metals) can exhibit more than one valency. These elements are found in the middle of the Periodic Table.

9 Use a textbook to give another definition for *transition metals*.

10 Give two examples of transition metals.

11 Look up the word *transition* in a dictionary. Now explain how you think the term *transition metal* came to be assigned to the transition metal elements.

12 Give your own definition for *valency*.

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Radicals

A **radical** is a charged group of atoms in compounds that always exists as a group.

The valency of a particular ion is often shown in Roman numerals inside brackets. Non-metals tend to form negative ions. Groups of non-metal atoms may combine to form **radicals**.

Some examples of radicals are sulphate $-SO_4^{2-}$, carbonate $-CO_3^{2-}$ and nitrate $-NO_3^{2-}$. Radicals have also been described as **polyatomic ions**. Table 2 shows some common ions, some of which are radicals.

–1 valency	–2 valency	–3 valency
hydroxide, OH^-	carbonate, CO_3^{2-}	phosphate, PO_4^{3-}
nitrite, NO_2^-	sulfate, SO_4^{2-}	nitride, N^{3-}
nitrate, NO_3^-	sulfite, SO_3^{2-}	phosphide, P^{3-}
ethanoate, CH_3COO^-	hydrogenphosphate, HPO_4^{2-}	
hydrogencarbonate, HCO_3^-	dichromate, $Cr_2O_7^{2-}$	
hydrogensulfate, HSO_4^-	chromate, CrO_4^{2-}	
hypochlorite, ClO^-	oxide, O^{2-}	
permanganate (VII), MnO_4^-	sulfide, S^{2-}	
cyanide, CN^-	oxalate $C_2O_4^{2-}$	
hydride, H^-		
chloride, Cl^-		
bromide, Br^-		
fluoride, F^-		
iodide, I^-		

Table 2

13 State two things that the ions in table 2 have in common.

14 Write the formula for each ion.

- **a** hydrogen phosphate
- **b** hydrogencarbonate
- **c** sulfate
- **d** chromate

15 Give the names and numbers of atoms present in each ion.

- **a** oxalate
- **b** phosphate
- **c** carbonate

16 Define a *radical*.

17 Give an example of a substance that contains a radical.

18 Give the type of charge that is likely to be found on ions of:

- **a** metals
- **b** non-metals

19 From Table 2, give the names and symbols for two ions that are not radicals according to the definition in the grey box.

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Formulae of ionic compounds

The formula of an ionic compound represents the relative numbers of each ion present. Formulae can be worked out by balancing the charges of the ions. For example, in calcium chloride the calcium ion (Ca^{2+}) has a charge of 2+ and the chloride ion (Cl^-) has a charge of 1–. Thus, the simplest unit of calcium chloride requires one calcium ion to balance two chloride ions. Therefore the formula of calcium chloride is $CaCl_2$.

20 Work out the formula for each compound.

- **a** sodium hydroxide
- **b** calcium hydroxide
- **c** magnesium oxide
- **d** copper (I) chloride
- **e** copper (II) chloride
- **f** ammonium carbonate
- **g** iron (III) oxide
- **h** barium nitrate
- **i** copper (II) sulfate
- **j** nickel (II) phosphate
- **k** lead (IV) sulfide
- **l** potassium chromate
- **m** barium phosphide
- **n** lithium cyanide
- **o** aluminium oxide

21 Copy and complete the table. The first one has been done for you.

Common name	Scientific name	Formula
table salt	sodium chloride	NaCl
		$NaHCO_3$
	calcium hydroxide	$Ca(OH)_2$
gypsum	calcium sulfate	
epsom salts		$MgSO_4$
Condy's crystals	potassium permanganate	
Glauber's salt		Na_2SO_4
caustic soda	sodium hydroxide	
saltpetre		KNO_3
potash		K_2CO_3

22 Give the name and formula for the radicals present in each substance.

- **a** epsom salts
- **b** saltpetre

23 Mark each statement true or false.

- **a** The ammonium ion contains metals.
- **b** Valency is the number of atoms from an atom involved in bonding.
- **c** Transition metals can have more than one valency.
- **d** Radicals only contain one type of atom.
- **e** Radicals act much like a single (large) atom in reactions.
- **f** The overall electrical charge in an ionic compound is zero.
- **g** In the name of a compound containing a metal, the name of the metal comes last.
- **h** The common name of a chemical usually tells you something about the formula.

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